

Shannova Technical Update Log

Date: March 25, 2026

Time: 23:06 PM

1. Executive Summary

This document outlines the critical bug fixes, infrastructure optimizations, and major feature additions applied to the Shannova Platform since the initial deployment.

Key Achievements:

- Solved severe cold-start backend timeouts that crashed the upload flow.
- Added advanced Statistical & Information Theory analytics (Shannon Entropy) to the AI pipeline.
- Added an automated GDPR Privacy Report that counts PII redactions.
- Updated the Frontend UI to elegantly display datasets with "Try before you buy" data samples.

2. Infrastructure & Cost Optimization Fixes

A. The Server Cold-Start Fix (Pre-Baking AI)

- **The Problem:** When uploading a file, the `curl` request (and browser) timed out with an `HTTP2 framing layer` error. This happened because Google Cloud Run spun up an empty container and spent 3 minutes attempting to download 1.5GB of Hugging Face models (`DeBERTa` and `i2b2`) over the network before it could process the file.
- **The Fix:** We injected a `download_models.py` script directly into the backend `Dockerfile`. Now, the models are downloaded *during* the build phase and "baked" permanently into the hard drive of the Docker image.
- **Result:** Uploads are now processed instantly without network download latency.

B. OOM (Out of Memory) Crash Fix

- **The Problem:** The backend was crashing because allocating Pandas Dataframes

3. Data Pipeline Analytics Upgrades

The backend `pipeline.py` was fundamentally rewritten to extract deep analytics from uploaded datasets before they reach the database.

A. Information Theory Metrics (Alpha Engine)

- The pipeline now uses `scipy.stats.entropy` to calculate the **Shannon Entropy** across categorical columns. This gives buyers a mathematical representation of the dataset's variance and unpredictability (i.e., higher entropy means richer information).
- It also mathematically calculates and stores the Exact Row Count, Column Count, and Missing Data Percentages.

B. The Privacy Report (Gamma Engine)

- The Gamma PII Eraser now generates a quantitative report.
- It compares the original DataFrame against the scrubbed DataFrame to count exactly

4. Frontend UI/UX Improvements

The Next.js React application was updated to visually display the new pipeline analytics and improve user navigation.

A. Navigation Renaming

- The ambiguous dropdown options under "Marketplace" were changed.
- "Upload" was renamed to **Selling** (the entry point for data providers).
- "Selling" was renamed to **Buying** (the entry point for data consumers).

B. The Dataset Details Page

The `/selling/[hash]/page.tsx` was heavily updated to include three new UI components:

1.  **Privacy & PII Report:** Highlights the Gamma engine's redaction count in red or green